

1 Input Variables, Prediction Target and Preprocessing

1.1 Input Variable

The study employs only a single input feature, namely the one-day stock return:

$$R_t^{1,s} = \frac{P_t^s}{P_{t-1}^s} - 1 \quad (1)$$

where P_t^s denotes the price of stock s at time t .

This quantity represents the daily percentage change in the stock price relative to the previous trading day.

1.2 Prediction Target

Rather than forecasting future stock prices directly, the paper formulates the task as a binary classification problem.

For each trading day:

1. The future return of every stock is computed.
2. The cross-sectional median return is calculated.
3. Stocks whose future return exceeds the median are assigned to Class 1 (*Outperform*).
4. Stocks whose future return falls below the median are assigned to Class 0 (*Underperform*).

1.3 Preprocessing

The paper applies two primary preprocessing steps before training the LSTM model.

1.3.1 Standardization

The return series is standardized using the mean and standard deviation computed exclusively from the training data:

$$\tilde{R}_t = \frac{R_t - \mu_{\text{train}}}{\sigma_{\text{train}}} \quad (2)$$

where μ_{train} and σ_{train} denote the mean and standard deviation of the training dataset, respectively.

1.3.2 Sequence Construction

After standardization, the returns are transformed into overlapping sequences of length 240 using a sliding-window approach:

$$[\tilde{R}_1, \tilde{R}_2, \dots, \tilde{R}_{240}] \quad (3)$$

$$[\tilde{R}_2, \tilde{R}_3, \dots, \tilde{R}_{241}] \quad (4)$$

$$[\tilde{R}_3, \tilde{R}_4, \dots, \tilde{R}_{242}] \quad (5)$$

and so forth until the end of the dataset. Each sequence serves as one input sample for the LSTM network.

2 Training, Validation and Testing Periods

In Paper a rolling-window evaluation framework in which the complete dataset is divided into multiple *study periods*. Each study period consists of a training phase followed by an out-of-sample trading (testing) phase.

2.1 Study Period Structure

Each study period contains:

- **Training Period:** 750 trading days (approximately 3 years)
- **Trading/Testing Period:** 250 trading days (approximately 1 year)
- **Total Length:** 1000 trading days

The entire dataset spanning December 1990–2015 January is partitioned into 23 non-overlapping study periods. Once a trading period ends, the next study period begins immediately after it, ensuring that no overlap exists between consecutive trading periods.

2.2 Rolling Window Procedure

For each study period, the following procedure is performed:

1. The model is trained using the first 750 trading days.
2. The trained model is then evaluated on the subsequent 250 trading days, which remain completely unseen during training.
3. After the trading period ends, the entire window is shifted forward by 250 trading days.
4. The training and testing process is repeated for the new study period.

2.3 Validation Split within the Training Period

The 750-day training period is further divided into two subsets:

- **80% Training Set** – used for model parameter learning.
- **20% Validation Set** – used for monitoring performance and implementing early stopping.

3 Baseline Models, LSTM Specification and Evaluation Metrics

3.1 Baseline Models

To evaluate the effectiveness of the proposed LSTM model, the study compares its performance against three widely used benchmark models:

1. **Logistic Regression (LR)** – a simple linear classification model commonly used as a baseline in financial prediction tasks.
2. **Random Forest (RF)** – an ensemble learning method consisting of 1000 decision trees with a maximum tree depth of 20.

3. **Deep Neural Network (DNN)** – a feed-forward neural network with the following architecture:

$$31 \rightarrow 31 \rightarrow 10 \rightarrow 5 \rightarrow 2 \quad (6)$$

where the final layer contains two output neurons corresponding to the two prediction classes.

3.2 LSTM Architecture

The proposed Long Short-Term Memory (LSTM) network is specifically designed to model temporal dependencies in stock return sequences.

The architecture consists of:

- **Input Features:** One standardized return feature
- **Sequence Length:** 240 trading days
- **LSTM Hidden Units:** 25
- **Dropout Rate:** 0.1
- **Output Layer:** Two neurons with Softmax activation
- **Total Trainable Parameters:** 2752

3.3 Training Configuration

The LSTM network is trained using the following configuration:

- **Optimizer:** RMSprop
- **Loss Function:** Cross-Entropy Loss
- **Maximum Epochs:** 1000
- **Early Stopping Patience:** 10 epochs

3.4 Trading Strategy

The trained model outputs the probability that each stock will outperform the market median return.

At each trading date:

1. Stocks are ranked according to their predicted probability of outperforming.
2. The **Top 10** ranked stocks are selected for long positions.
3. The **Bottom 10** ranked stocks are selected for short positions.

This produces a market-neutral long-short portfolio consisting of twenty stocks.

3.5 Evaluation Metrics

The study evaluates model performance from both a predictive and financial perspective.

3.5.1 Predictive Performance Metrics

The following statistical measures are used to assess prediction quality:

- **Accuracy** – percentage of correctly classified observations.
- **Diebold-Mariano (DM) Test** – evaluates whether the forecasting performance of two competing models is significantly different.
- **Pesaran-Timmermann (PT) Test** – examines whether directional predictions are significantly better than random guessing.

3.5.2 Financial Performance Metrics

The profitability and risk characteristics of the trading strategy are evaluated using:

- **Mean Daily Return**
- **Sharpe Ratio**

$$\text{Sharpe Ratio} = \frac{\mu_p - r_f}{\sigma_p} \quad (7)$$

- **Sortino Ratio**

$$\text{Sortino Ratio} = \frac{\mu_p - r_f}{\sigma_d} \quad (8)$$

- **Standard Deviation**

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (R_i - \bar{R})^2} \quad (9)$$

- **Maximum Drawdown (MDD)**

$$\text{MDD} = \frac{\text{Peak Value} - \text{Trough Value}}{\text{Peak Value}} \quad (10)$$

- **Value at Risk (VaR)**

4 Work Completed, Difficulties Encountered and Next Steps

4.1 Work Completed

- Regarding the dataset, I contacted Harikrishna Sir and requested the required data. He has agreed to provide the dataset as soon as possible.
- I have read the research paper multiple times (approximately two to three complete readings) to gain a thorough understanding of the methodology and experimental setup.
- Over the coming week, I plan to revisit the paper in greater depth, particularly focusing on the financial concepts and terminology used throughout the study, as some aspects still require further clarification.

4.2 Difficulties Encountered

4.2.1 Dataset-Related Challenges

One of the primary challenges encountered during this phase was understanding the structure and requirements of the dataset used in the study. The dataset comprises multiple components, including historical stock price data, S&P 500 constituent information, company-level characteristics, corporate actions, benchmark market data, and valuation variables.

4.2.2 Financial Terminology

I also faced some difficulty understanding a number of financial concepts and evaluation metrics used in the paper. As several of these terms were new to me.

4.3 Next Steps

1. Develop a comprehensive understanding of all financial terminology and evaluation metrics used in the paper.
2. Perform a detailed review of the methodology to ensure complete clarity regarding data preparation, model training, and evaluation procedures.
3. Finalize the implementation structure, including notebooks, folder organization, and reporting framework and start to implement.